MTH 2215
Applied discrete mathematics

Chapter 2, Section 2.1
Sets

These class notes are based on material from our textbook, **Discrete Mathematics and Its Applications**, 6th ed., by Kenneth H. Rosen, published by McGraw Hill, Boston, MA, 2006. They are intended for classroom use only and are **not** a substitute for reading the textbook.

Sets

- A *set* is an unordered collection of objects.
- The set is the fundamental discrete structure on which all other discrete structures are built.
- The objects in a set are called its *elements*, or *members*.
- A set is said to *contain* its elements.

Set Notation

- We normally use upper-case letters to represent the names of sets, and lower-case letters to represent their elements.
- To denote that *a* is an element of set *S* we write: $a \in S$
- To denote that *a* is not an element of set *S* we write: $a \notin S$

How to Describe a Set

- We can describe a set in two ways:
  - List all of its elements
  - Give a set of rules that characterize all of the members of the set (*set builder* notation)

Listing the Elements of a Set

- To list the members of a set, we use curly braces, separating each element from the next with a comma.
- Example: the set of all vowels in the English language is the set

  $V = \{a, e, i, o, u\}$
- We can use ellipses to keep us from having to list all of the elements individually, provided the meaning is obvious: $H = \{1, 2, 3, 4, \ldots, 100\}$

Using Set Builder Notation

- Often we are dealing with sets where it is impossible to list all of their elements.
- In set builder notation, we give a rule that characterizes all members of a set.
- Example:

$$S = \{x \mid x \text{ is the square of an integer}\}$$

- This can be read, “S is the set of all x such that x is the square of an integer”.

Using Set Builder Notation

- In studying computer theory, we find it useful to remember the following sets:

$N = \{0, 1, 2, 3, \ldots\}$, the set of *natural numbers*

$Z = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$, the set of *integers*

$Z^+ = \{1, 2, 3, \ldots\}$, the set of *positive integers*

$Q = \{p/q \mid p \in Z, q \in Z, \text{ and } q \neq 0\}$, the set of *rational numbers*

Q^+ = the set of *positive rational numbers*

R = the set of *real numbers*

What Can Constitute a Set?

- Note that anything within a set of curly braces can be considered a set. The elements of a set don't necessarily have to have anything to do with one another.
- Example: {Boston, "7", iPod, 2.7, Sleepy} is a legal set.

What Can Constitute a Set?

- Two sets are *equal* if and only if they have the same elements.
- Consider sets A and B. Then A = B (A and B are equal) iff:

$$\forall x ((x \in A) \leftrightarrow (x \in B))$$

The Elements of a Set

The order in which elements occur in sets is irrelevant. For example, the following two sets are equal:

{a, b, c, d, e}

{c, e, a, d, b}

The Elements of a Set

It does not matter if an element of a set is listed more than once. For example,

{a, a, a, b, c}

and

{a, b, c}

are equivalent. We ignore any duplicates.

What Can Constitute a Set?

- The elements of a set can themselves be sets. For example,

  $S = \{N, Z, Q, R\}$

- Question: Does set S have any duplicate elements?
- Answer: No. Set S has only 4 elements, none of which is equivalent to any of the others.

What Can Constitute a Set?

- A set that has no elements is called the *empty set* or *null set*.
- Yes, it is still considered a real set, even though it has no elements.
- It is denoted by ∅, or by { }.
- Since the empty set is a set, another set can contain the empty set as one of its elements:

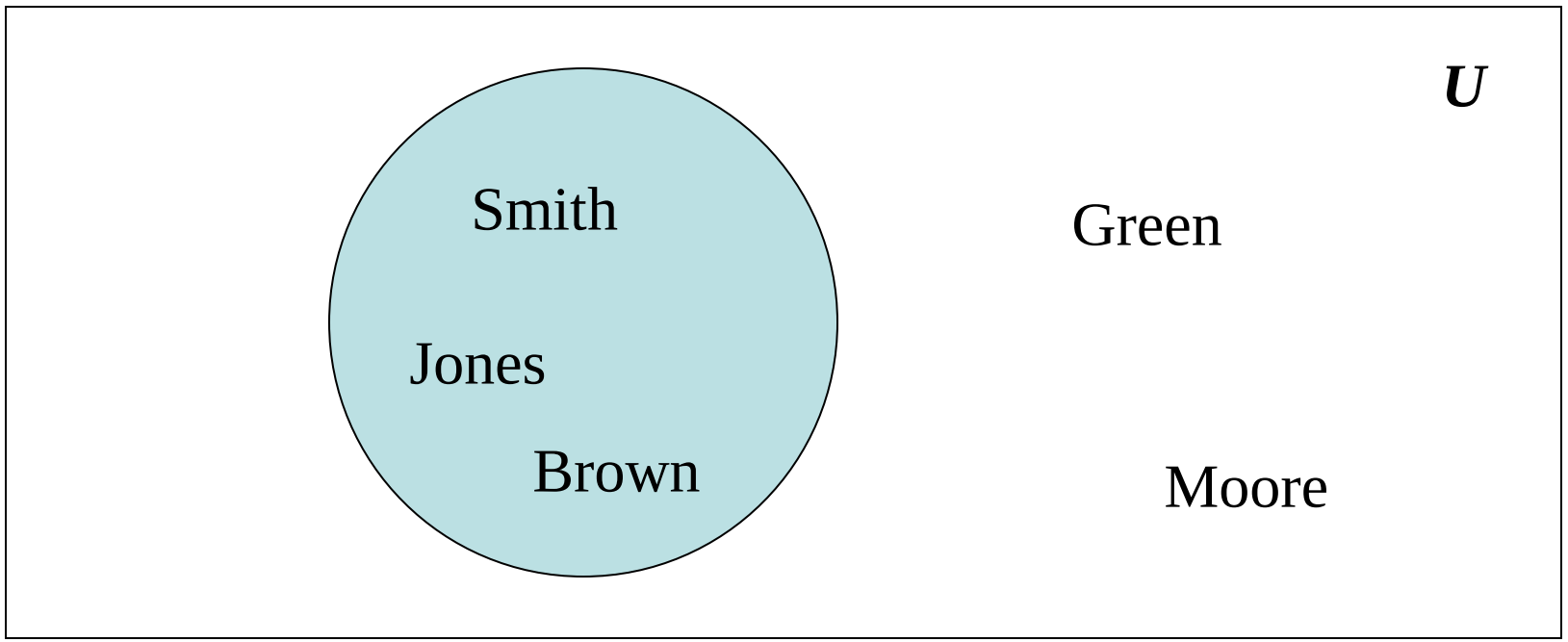
A ={∅, a}	This set has 2 elements
B = {∅}	This set has 1 element
C = ∅	This set has 0 elements

Venn Diagrams

- Sets can be represented graphically using Venn diagrams.
- In Venn diagrams:
  - A rectangle represents the universal set (universe of discourse)
  - Circles (and other geometric figures) represents sets
  - Points (or words) represent elements

Venn Diagrams

- Assistant professors at Blivet State University who have taught MTH 2215:



Subset

The set A is said to be a *subset* of set B if and only if every element of set A is also an element of set B.

We use this notation: $A \subseteq B$

$A \subseteq B$ is true if and only if the following quantification is true:

$$\forall x ((x \in A) \rightarrow (x \in B))$$

Subset

Obviously, according to the preceding definition, if A = B, it must be true that:

$A \subseteq B$, and

$B \subseteq A$.

Moreover, it should be self-evident that every set is a subset of itself. That is:

$A \subseteq A$

Proper Subset

However, if all of A's elements are also in B, *but* B has some elements in it that A does not have (that is, $A \neq B$), then we can be a more precise and say that A is a *proper subset* of B.

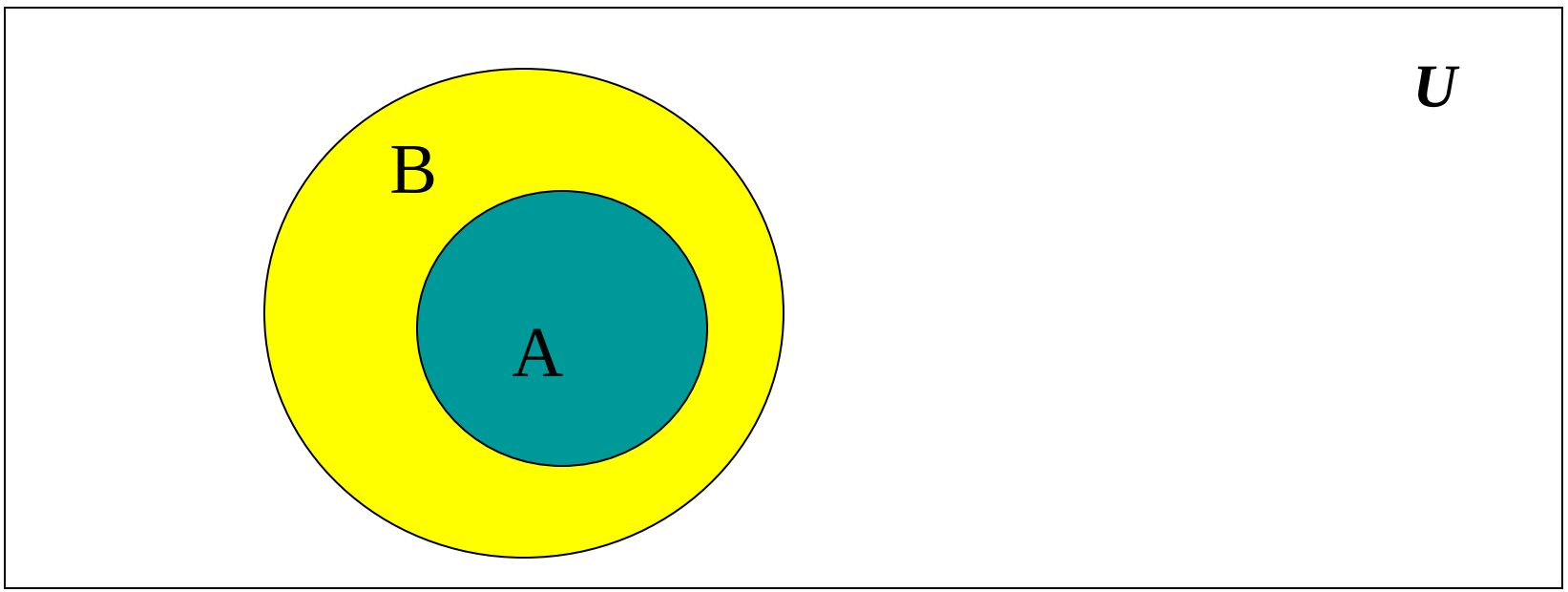
We use this notation: $A \subset B$

A is a proper subset of B iff:

$$\forall x\,(x \in A \rightarrow x \in B) \wedge \exists x(x \in B \wedge x \notin A)$$

Proper Subset

We can represent the subset relationship using a Venn diagram. The following diagram represents $A \subset B$:



Subset

Interestingly enough, the empty set, ∅, is a subset of every other set (or, more precisely, every *nonempty* set).

Your book gives a formal proof, but you might think of it this way instead: if set S = {a, b}, then it has 4 subsets –

{{a}, {b}, {a, b}, ∅}

Properties of Sets

One way to show that two sets are equal is to show that each set is a subset of the other.

The Cardinality of a Set

Given a set S, and $n \in \mathbb{N}$ (that is, n is an element of the set of natural numbers -- the integers from 0 on up),

if there are exactly n distinct elements in S, then:

S is a *finite set*, and

n is the *cardinality* of S

The cardinality of S is represented by |S|.

Properties of Sets

We now can see that every nonempty set S must have at least two subsets:

$\varnothing$ and S

Theorem 1 in section 2.1 of your textbook says:

For every set S,

$\varnothing \subseteq S$

$S \subseteq S$

Powerset

The *powerset* of S is the set of all subsets of S.

The powerset of S is represented by $P(S)$, or by the symbol 2^S

For example, if S = {a, b}, then:

$P(S) = \{\{a\}, \{b\}, \{a, b\}, \varnothing\}$

Powerset

Remember that |S| represents the cardinality of S (the number of elements in S).

Here S has two elements, *a* and *b*. So $2^{|S|}$ can be understood as 2^2, which is 4.

And 4 is the number of subsets of S, or the *cardinality of the powerset* of S.

Powerset

The powerset of the empty set is a special case.

The powerset of the empty set is:

$P(\varnothing) = \{\varnothing\}$

Note $P(\{\varnothing\}) = \{\varnothing, \{\varnothing\}\}$

Cartesian Product

The Cartesian product, or cross product, of two sets is the set of *ordered pairs* of elements of the two sets. To represent the cross product of sets A and B we use the symbol ×, as in A × B.

For example, given:

set A = {a, b} and

set Y = {x, y}

The Cartesian product A × Y = {(a,x), (a,y), (b,x), (b,y)}

Set Notation with Quantifiers

$\forall x \in S(P(x))$ means " for all x that are elements of S, P(x) is true". This is referred to as the *universal quantification of P(x) over all elements in the set S.* It is shorthand for: $\forall x(x \in S \rightarrow P(x))$

$\exists\ x \in S(P(x))$ is the *existential quantification of P(x) over all elements in the set S.* It is shorthand for: $\exists x(x \in S \wedge P(x))$

Truth Sets of Quantifiers

Given a predicate, P, and a domain, D, the *truth set* of P is defined as the set of elements in D for which P(x) is true.

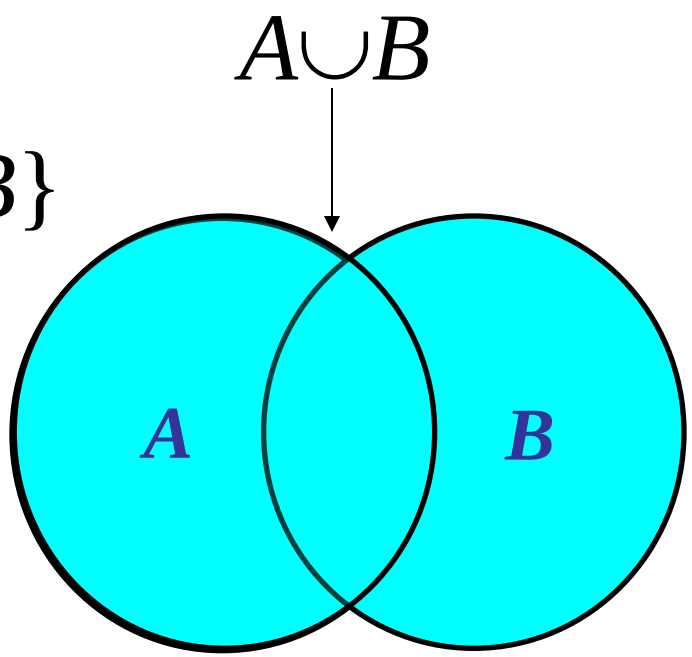
The truth set of P(x) is is denoted by $\{x \in D \mid P(x)\}$: “those elements of domain D such that P(x) is true”

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Chapter 2, Section 2.2
Set Operations

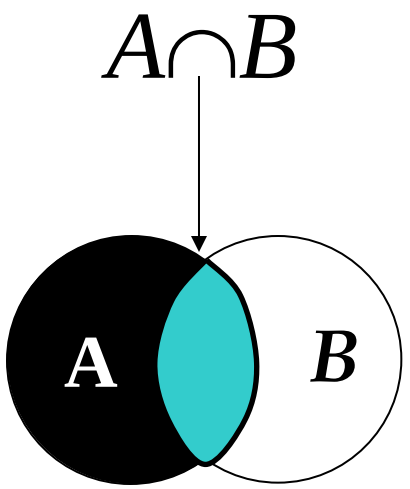
Set Union

- Union of two sets A and B is denoted by $A \cup B$
- $A \cup B$ contains elements that are either in A or in B or in both.
- $A \cup B = \{x \mid x \in A \vee x \in B\}$
- $A = \{1,3,5\}$, $B = \{2,3,4\}$
- $A \cup B = \{1, 2, 3, 4, 5\}$



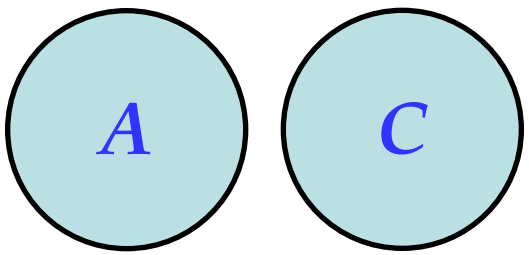
Set Intersection

- Intersection of two sets A and B is denoted by $A \cap B$
- $A \cap B$ contains elements that are in both A and B
- $A \cap B = \{x \mid x \in A \wedge x \in B\}$
- $A = \{1,3,5\}$, $B = \{1,2,3\}$
- $A \cap B = \{1, 3\}$



Disjoint Sets

- Two sets are called disjoint if their intersection is the empty set.
- $A = \{1,3,5\}$, $B = \{1,2,3\}$, $C = \{6,7,8\}$
- Are A and B disjoint? NO
- Are A and C are disjoint? YES



Cardinality of the Union of Sets

How many elements does A∪B have?

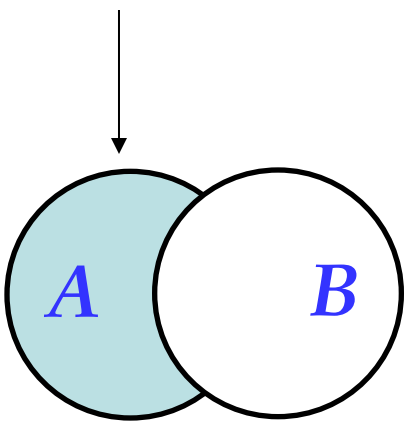
The number of elements in A plus the number of elements in B, minus the number of elements in both sets.

This can be written:

$$|A \cup B| = |A| + |B| - |A \cap B|$$

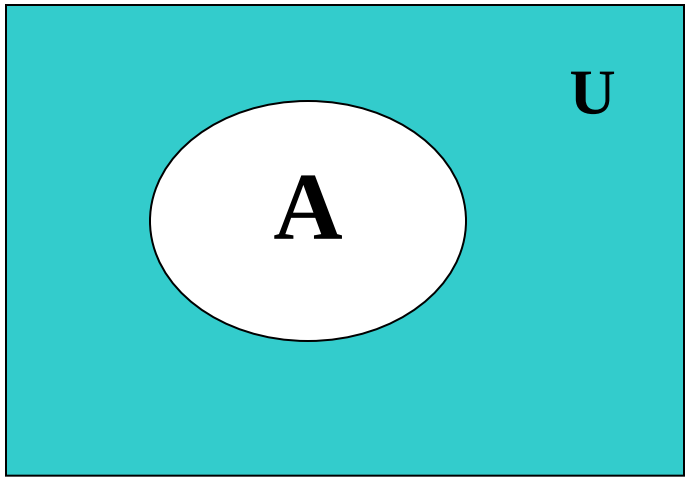
Set Difference

- Difference of two sets A and B is denoted by $A–B$
- $A–B$ contains elements that are in A but not in B.
- $A–B = \{x \mid x \in A \wedge x \notin B\}$
- $A = \{1,3,5\}$, $B = \{1,2,3\}$
- $A–B = \{5\}$



Complement of a Set

- Complement of a set A is denoted by $\overline{A}$
- Done with respect to a Universal set U
- $\overline{A}$ contains elements which are not in A, but are in U.
- $\overline{A} = U - A$
- $\overline{A} = \{x \mid x \in U \wedge x \notin A\}$



Set Identities

$A \cup \varnothing = A$	Identity
$A \cap U = A$	
$A \cup U = U$	Domination
$A \cap \varnothing = \varnothing$	
$A \cup A = A$	Idempotent
$A \cap A = A$	
$\overline{\overline{A}} = A$	Double Complement

Set Identities (Cont.)

$A \cup B = B \cup A$	Commutative
$A \cap B = B \cap A$	
$A \cup (B \cup C) = (A \cup B) \cup C$	Associative
$A \cap (B \cap C) = (A \cap B) \cap C$	
$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$	Distributive
$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$	
$A \cup (A \cap B) = A$	Absorption
$A \cap (A \cup B) = A$	
$\overline{A \cup B} = \overline{A} \cap \overline{B}$	De Morgan's
$\overline{A \cap B} = \overline{A} \cup \overline{B}$	

Examples

- Use set builder notation to prove that:

$$\overline{A \cap B} = \overline{A} \cup \overline{B}$$

- Use set identities to prove that

$$\overline{\overline{(A \cup B)} \cap C} \cup \overline{B} = B \cap C$$

More Exercises

- Describe the following sets using the set builder notation:
  1. The set of all positive integers between 1 and 99.
  2. $\overline{A \cup B}$
  3. $\overline{A} \cap \overline{B}$
  4. $A \oplus B$
  5. $(A \cup B) \cap C$
- Use set builder notation to prove .

  $\overline{A \cup B} = \overline{A} \cap \overline{B}$

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Chapter 2, Section 2.3
Functions

Recap 2.1

- Set – an **UNORDERED** collection of objects
  - *Element /member* **-** an object in a set
  - Notation **-** {a,b,c,d}
- Cardinality
  - The number of distinct elements in a set
- Power Set
  - The set of all subsets of a set
- Cartesian product of two sets A and B $A \times B$

$$A \times B = \{(a, b) \mid a \in A \wedge b \in B\}$$

Recap 2.2

- Union: $A \cup B = \{x \mid x \in A \vee x \in B\}$
- Intersection: $A \cap B = \{x \mid x \in A \wedge x \in B\}$
- Difference: $A - B = \{x \mid x \in A \wedge x \notin B\}$
- Complement: $\bar{A} = U - A$
- Identities similar to those from logic, e.g.

Definitions

- Let A and B be sets. A *function* f from A to B is an assignment of *exactly* one element of B to *each* element of A.
- We write $\boldsymbol{f(a) = b}$ if b is the unique element of B assigned by the function f to the element of A.
- If $\boldsymbol{f}$ is a function from $\boldsymbol{A}$ to $\boldsymbol{B}$, we write

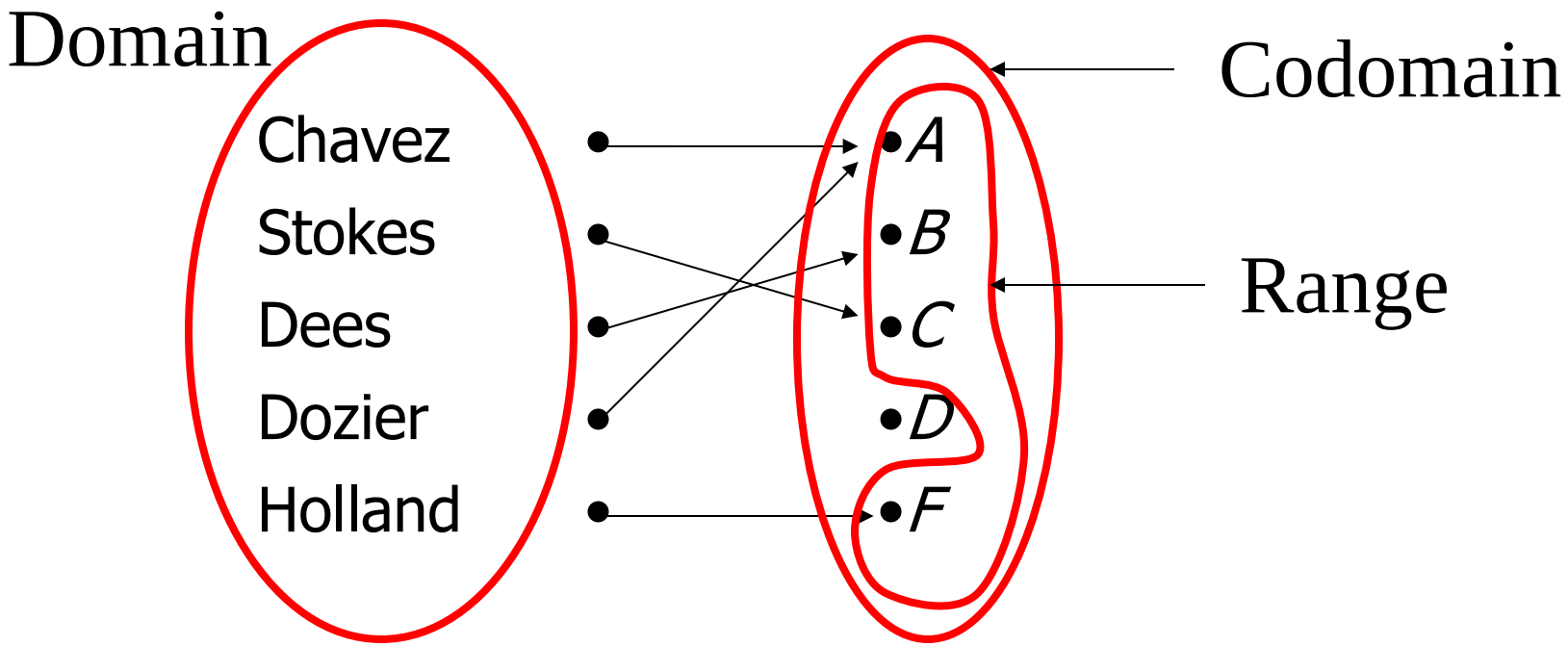
$$\boldsymbol{f : A \rightarrow B}$$

Definitions

- If $\boldsymbol{f : A \rightarrow B}$, we say that *A* is the *domain* of *f* and *B* is the *codomain* of *f*.
- If $\boldsymbol{f(a) = b}$, we say that *b* is the *image* of *a*.
- The *range* of *f* is the set of all images of elements of *A*.

Example

- Suppose that each student in a class is assigned a letter grade from the set $\{A, B, C, D, F\}$. Let g be the function that assigns a grade to a student.



Example

- Consider a function $f : \mathbf{Z} \rightarrow \mathbf{Z}$ that assigns the square of an integer to this integer.
- How can you write this function?

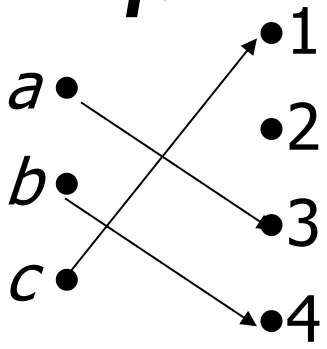
$$f(x) = x^2$$

- What is the domain of f ? The integers
- What is the codomain of f ? The integers
- What is the range of f ?

The nonnegative integers {0,1,4,9,..}

One-to-One Functions (injective)

- No value in the range is used by more than one value in the domain.
- If $f(x) = f(y)$, then $x = y$ for all x and y in the domain of f.



- In other words $\forall x\ \forall y\ (f(x) = f(y) \rightarrow x = y)$,

  or using the contrapositive

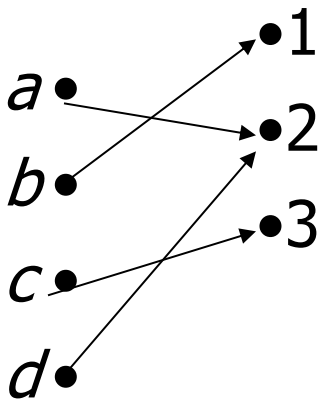
$$\forall x\ \forall y\ (x \neq y \rightarrow f(x) \neq f(y))$$

One-to-One Functions

- Is the function $\boldsymbol{f(x) = x^2}$ from the set of integers to the set of integers one-to-one?
  - $\forall x \, \forall y \, (x^2 = y^2 \rightarrow x = y)$?
  - $1^2 = (-1)^2$ but $1 \neq -1$
  - NO
- Is the function $\boldsymbol{f(x) = x + 1}$ one-to-one?
  - $\forall x \, \forall y \, (x + 1 = y + 1 \rightarrow x = y)$?
  - $(x + 1) \neq (y + 1)$ only when $x \neq y$
  - YES

Onto Functions (surjective)

- For every value in the codomain, there is a value in the domain that is mapped to it.



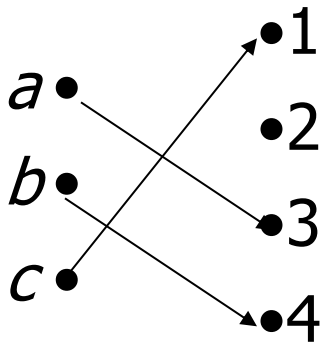
- In other words, $\forall y \exists x (f(x) = y)$
- Codomain = range!

Onto Functions

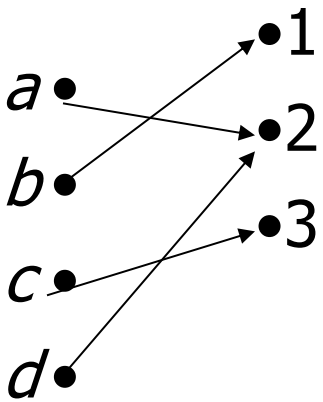
- Is the function $\boldsymbol{f(x) = x^2}$ from the set of integers to the set of integers onto?
  - Is it true that $\forall y \, \exists x \, (x^2 = y)$?
  - -1 is one of the possible values of y, but there does not exists an x such that $x^2 = -1$
  - NO
- Is the function $\boldsymbol{f(x) = x + 1}$ onto?
  - Is it true that $\forall y \, \exists x \, (x + 1 = y)$?
  - For every y, some x exists such that $x = y - 1$.
  - YES

One-to-One Correspondence (bijection)

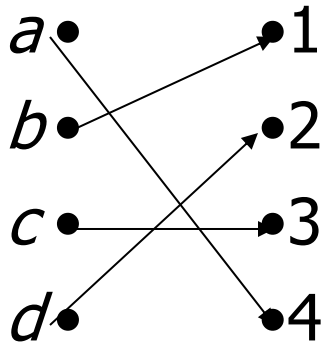
- If a function ***f*** is both one-to-one and onto, then it is a one-to-one correspondence.



One-to-One
but not Onto



Onto, but
Not One-to-One



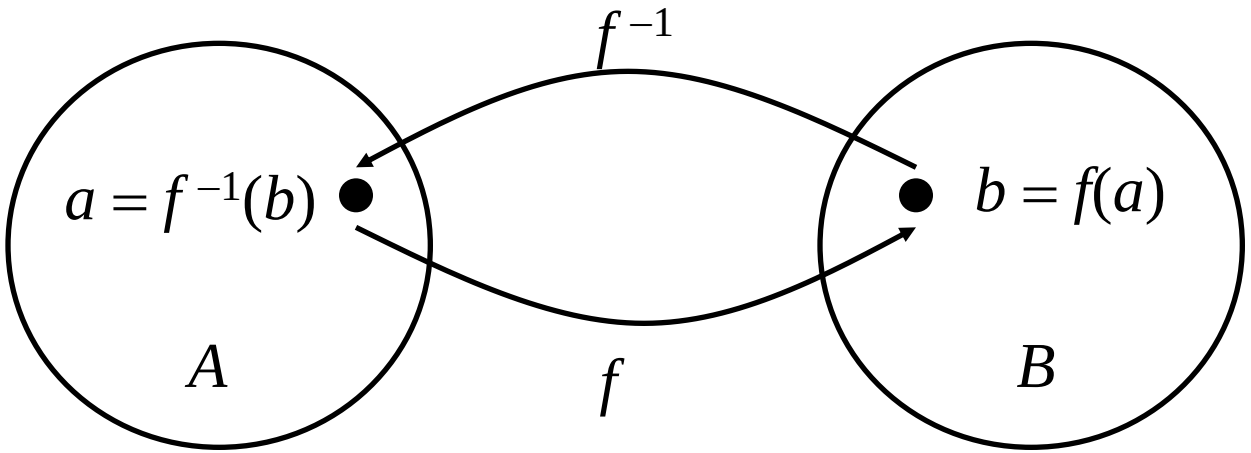
One-to-One
Correspondence

Monotonic Functions

- A monotonic function is
  - either *monotonically* (*strictly*) *increasing*
  - or *monotonically* (*strictly*) *decreasing*
- Consider a function $f : \mathbf{R} \rightarrow \mathbf{R}$
- f is monotonically increasing
  - if $f(x) \leq f(y)$ whenever $x < y$
- f is monotonically decreasing
  - if $f(x) \geq f(y)$ whenever $x < y$

Inverse Functions

- Let $\boldsymbol{f : A \rightarrow B}$ be <u>one-to-one correspondence</u> such that $\boldsymbol{f(a) = b}$.
- The inverse of the function $\boldsymbol{f}$ is denoted by $\boldsymbol{f^{-1}(b) = a}$.



F needs to be bijection

- If *f* is not a bijection (not one-to-one correspondence), then
  - *f* is not injective (not one-to-one)
  - *f* is not surjective (not onto)
- Why can't we invert such a function?

  We cannot assign to each element *b* in the codomain a unique element *a* in the domain such that $f(a) = b$, because:
  - For some *b* there is either
    - More than one *a*
    - No such *a*

Inverse Functions

- Let $f : \mathbf{Z} \rightarrow \mathbf{Z}$ be a function with $f(x) = x + 1$
- Is *f* invertible? Is *f* a bijection?
  - Is *f* one-to-one? YES
  - Is *f* onto? YES
  - So *f* is a one-to-one correspondence and is therefore invertible.
- Then, what is its inverse?

  f(y) = y - 1

Inverse Functions

- Let $f : \mathbf{Z} \to \mathbf{Z}$ be a function with $f(x) = x^2$.
- Is f invertible?
- Is f a one-to-one correspondence. NO
- So f is not a one-to-one, and
- therefore, f is not invertible.

Compositions of Functions

- Let $g : A \to B$ and $f : B \to C$.
- The *composition* of the functions f and g, denoted by $f \circ g$, is defined by:

$$f \circ g(a) = f(g(a))$$

- $f \circ g$ can't be defined unless the range of g is a subset of the domain of f.

Example

- Let:

  $f(x) = 2x + 3$

  $g(x) = 3x + 2$

- Find $f \circ g(x)$:

  $2(3x + 2) + 3$

- Find $g \circ f(x)$:

  $3(2x + 3) + 2$

Let:

$f(a) = b$, so

$f^{-1}(b) = a$

- Find $f^{-1} \circ f(a)$:

  a

- $f \circ f^{-1}(b)$

  b

Important functions – Floor

- Let ***x*** be a real number. The *floor function* is the closest integer less than or equal to ***x***.
- Examples:

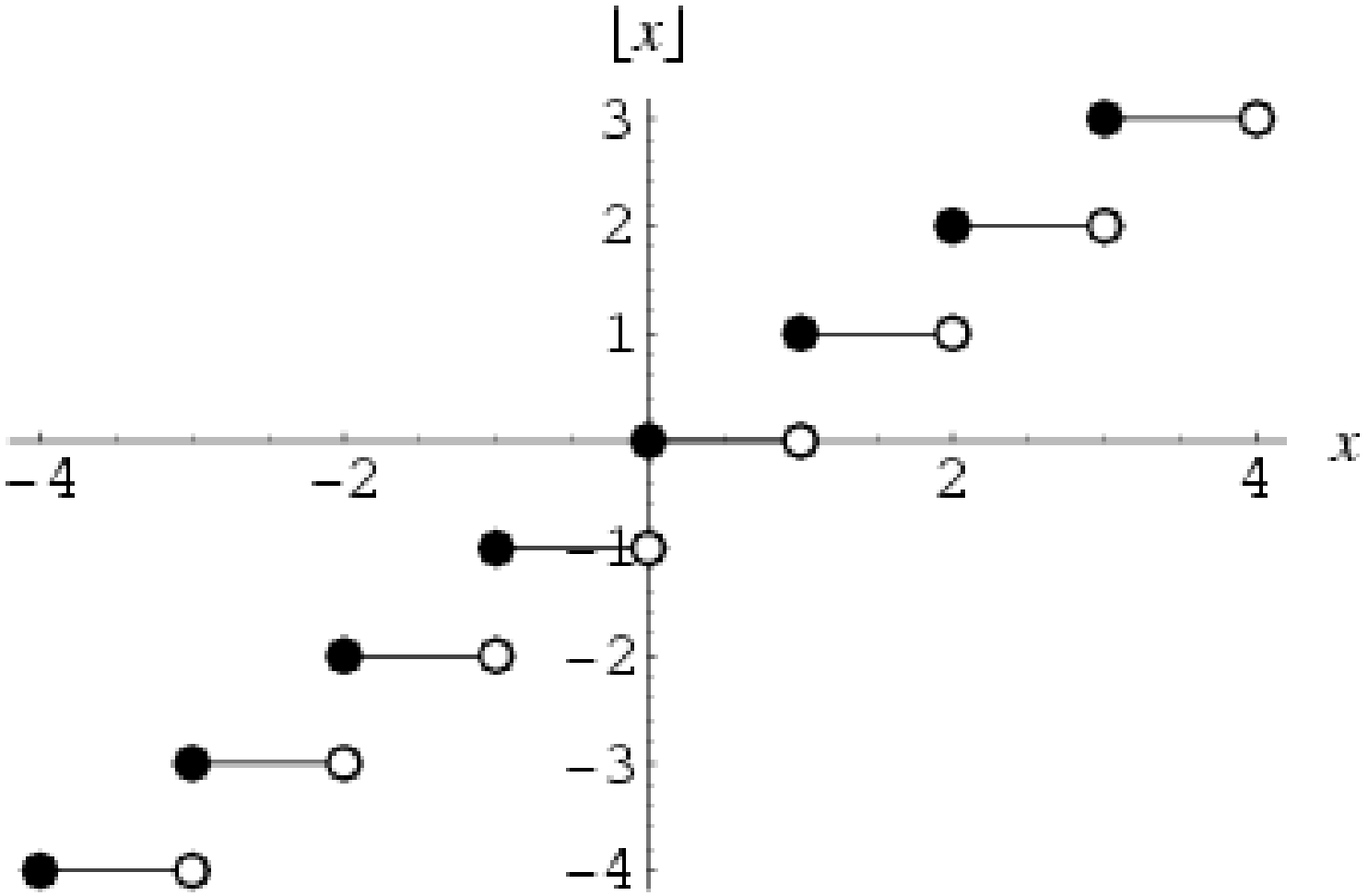
$$\lfloor \tfrac{1}{2} \rfloor = 0$$

$$\lfloor -\tfrac{1}{2} \rfloor = ?$$

$$\lfloor 3.1 \rfloor = ?$$

$$\lfloor 7 \rfloor = ?$$

Floor



http://mathworld.wolfram.com/FloorFunction.html

Important functions – Ceiling

- Let ***x*** be a real number. The *ceiling function* is the closest integer greater than or equal to ***x***.
- Examples:

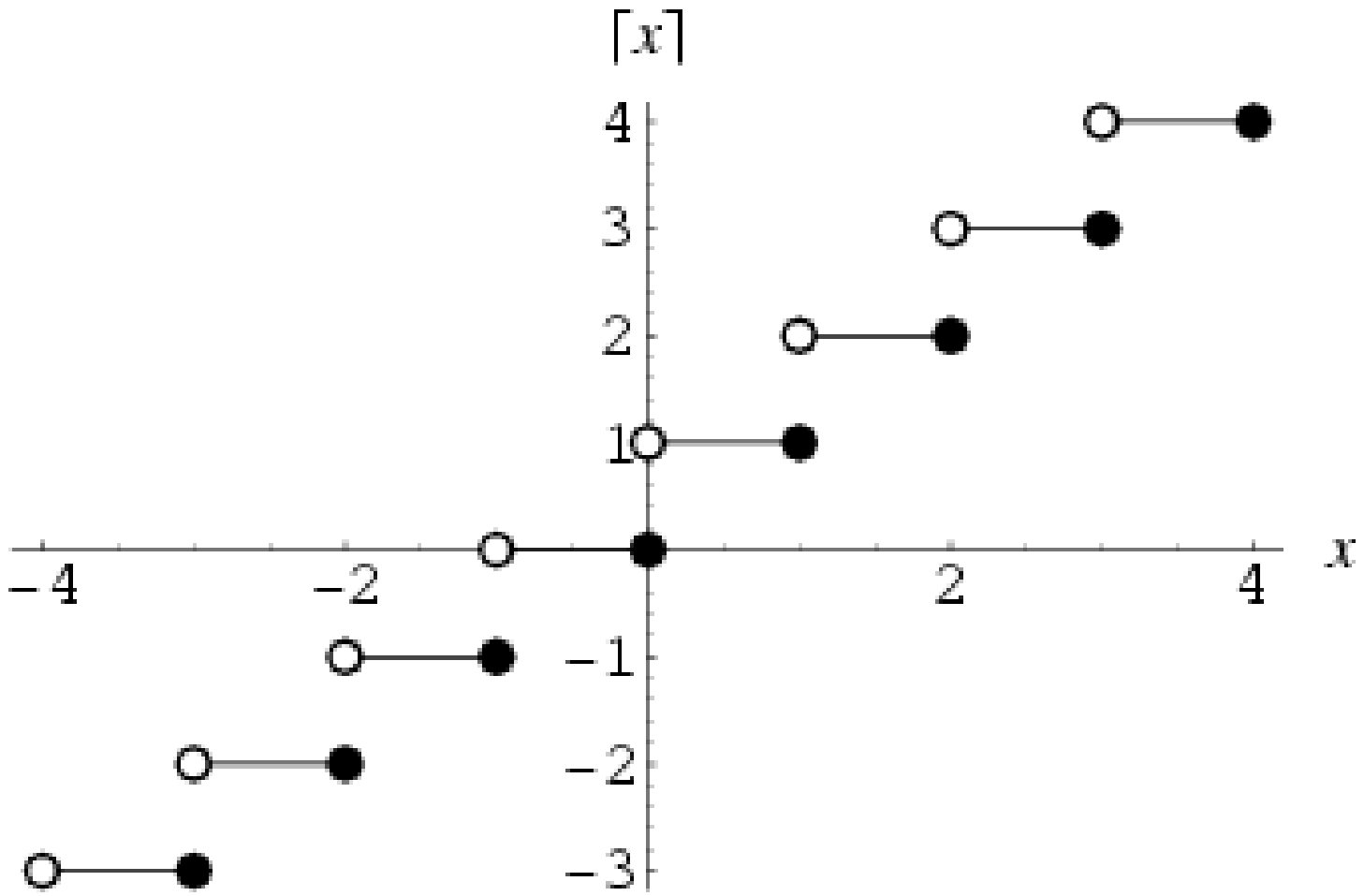
$$\lceil \tfrac{1}{2} \rceil = 1$$

$$\lceil -\tfrac{1}{2} \rceil = ?$$

$$\lceil 3.1 \rceil = ?$$

$$\lceil 7 \rceil = ?$$

Ceiling



http://mathworld.wolfram.com/CeilingFunction.html

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Chapter 2, Section 2.4
Sequences and Summations

Sequence

- If the domain of a function is restricted to integers, the function is called a *sequence*.
- The domain is specifically the set **N** or the set $\mathbf{Z}^+$.
- a_n denotes the image of n
  - called a *term* of the sequence
- Notation for whole sequence: $\{a_n\}$

Example

- Let $\{a_n\}$ be a sequence, where

  $a_n = 1/n$ and $n \in \mathbf{Z}^+$

- What are the *terms* of the sequence?

  $a_1 = 1$

  $a_2 = 1/2$

  $a_3 = 1/3$

  $a_4 = 1/4$

  … … …

Sequence Notation

- Unless stated to the contrary, we will assume the domain of a sequence to be the set of all positive integers.
- a_n is called the nth term or general term.

Geometric/Arithmetic Progression

- Geometric Progression:
  - A sequence of the form $a, ar, ar^2, \ldots, ar^n$
  - $a \in \mathbf{R}$ and $r \in \mathbf{R}$
  - a is the *initial term* and r is the *common ratio*
- Arithmetic Progression:
  - A sequence of the form $a, a+d, a+2d, \ldots, a+nd$
  - $a \in \mathbf{R}$ and $d \in \mathbf{R}$
  - a is the *initial term* and d is the *common difference*

Example

- Let $\{b_n\}$ be a sequence, where $b_n = (-1)^n$
  - What type of progression is this?
    (Geometric)
  - What is the initial term?
    (-1)
  - What is the common ratio/difference?
    (-1)
  - What are the terms of the sequence?
    (-1,1, -1, 1, -1, 1, …)

Example

- Let $\{d_n\}$ be a sequence, where $d_n = 6 \bullet (1/3)^n$
  - What type of progression is this?

    (Geometric)
  - What is the initial term?

    (2)
  - What is the common ratio/difference?

    (1/3)
  - What are the terms of the sequence?

    (2, 2/3, 2/9, 2/27, …)

Example

- Let $\{s_n\}$ be a sequence, where $s_n = -1 + 4n$
  - What type of progression is this?

    (Arithmetic)
  - What is the initial term?

    (3)
  - What is the common ratio/difference?

    (4)
  - What are the terms of the sequence?

    (3, 7, 11,15, …)

Example

- Let $\{t_n\}$ be a sequence, where $t_n = 7 - 3n$
  - What type of progression is this?

    (Arithmetic)
  - What is the initial term?

    (4)
  - What is the common ratio/difference?

    (-3)
  - What are the terms of the sequence?

    (4, 1, -2, -5, …)

Example

- Find a formula for this sequence:

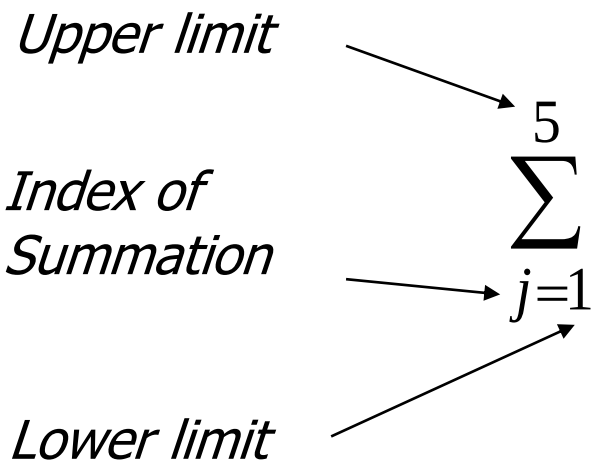
  1, 1/2, 1/3, 1/4, 1/5, ...

  What is the formula?

  ($a_n = 1/n$)

Summations

- A summation denotes the sum of the terms of a sequence.
- Example:



$$\sum_{j=1}^{5} j^2 = 1^2 + 2^2 + 3^2 + 4^2 + 5^2$$

$$= 1 + 4 + 9 + 16 + 25$$

$$= 55$$

Geometric Series

- The sum of a geometric progression is called a geometric series
- Commonly used

$$S = \sum_{j=0}^{n} ar^{j} = ar^{0} + ar^{1} + ar^{2} + \ldots + ar^{n}$$

Double Summation

$$\sum_{i=1}^{3}\sum_{j=1}^{2}(i-j)$$

$$\sum_{j=1}^{2}(i-j)=(i-1)+(i-2)=2i-3$$

$$\sum_{i=1}^{3}(2i-3)=(2\cdot1-3)+(2\cdot2-3)+(2\cdot3-3)$$

$$=-1+1+3=3$$

Useful Summation Formulae

TABLE 2 Some Useful Summation Formulae.

Sum	*Closed Form*
$\sum_{k=0}^{n} ar^k \ (r \neq 0)$	$\frac{ar^{n+1} - a}{r - 1}, r \neq 1$
$\sum_{k=1}^{n} k$	$\frac{n(n+1)}{2}$
$\sum_{k=1}^{n} k^2$	$\frac{n(n+1)(2n+1)}{6}$
$\sum_{k=1}^{n} k^3$	$\frac{n^2(n+1)^2}{4}$
$\sum_{k=0}^{\infty} x^k, \lvert x \rvert < 1$	$\frac{1}{1-x}$
$\sum_{k=1}^{\infty}, kx^{k-1}, \lvert x \rvert < 1$	$\frac{1}{(1-x)^2}$

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Cardinality of Infinite Sets

- A finite set is obviously countable. How about an infinite set?
- Remember that sets A and B have the same cardinality (number of elements) iff there is a one-to-one correspondence between A and B.
- We say that a set S is *countable* iff there is a one-to-one correspondence between S and Z^+, the set of positive integers.
- A set that is not countable is called *uncountable*.

Cardinality of Infinite Sets

- The set of all integers is countable.
- This means that there is a one-to-one correspondence between the set of *all* integers and the set of *positive* integers.
- We establish the one-to-one correspondence as follows:

…	***11***	***9***	***7***	***5***	***3***	***1***	***2***	***4***	***6***	***8***	***10…***
…	**-5**	**-4**	**-3**	**-2**	**-1**	**0**	**1**	**2**	**3**	**4**	**5** …

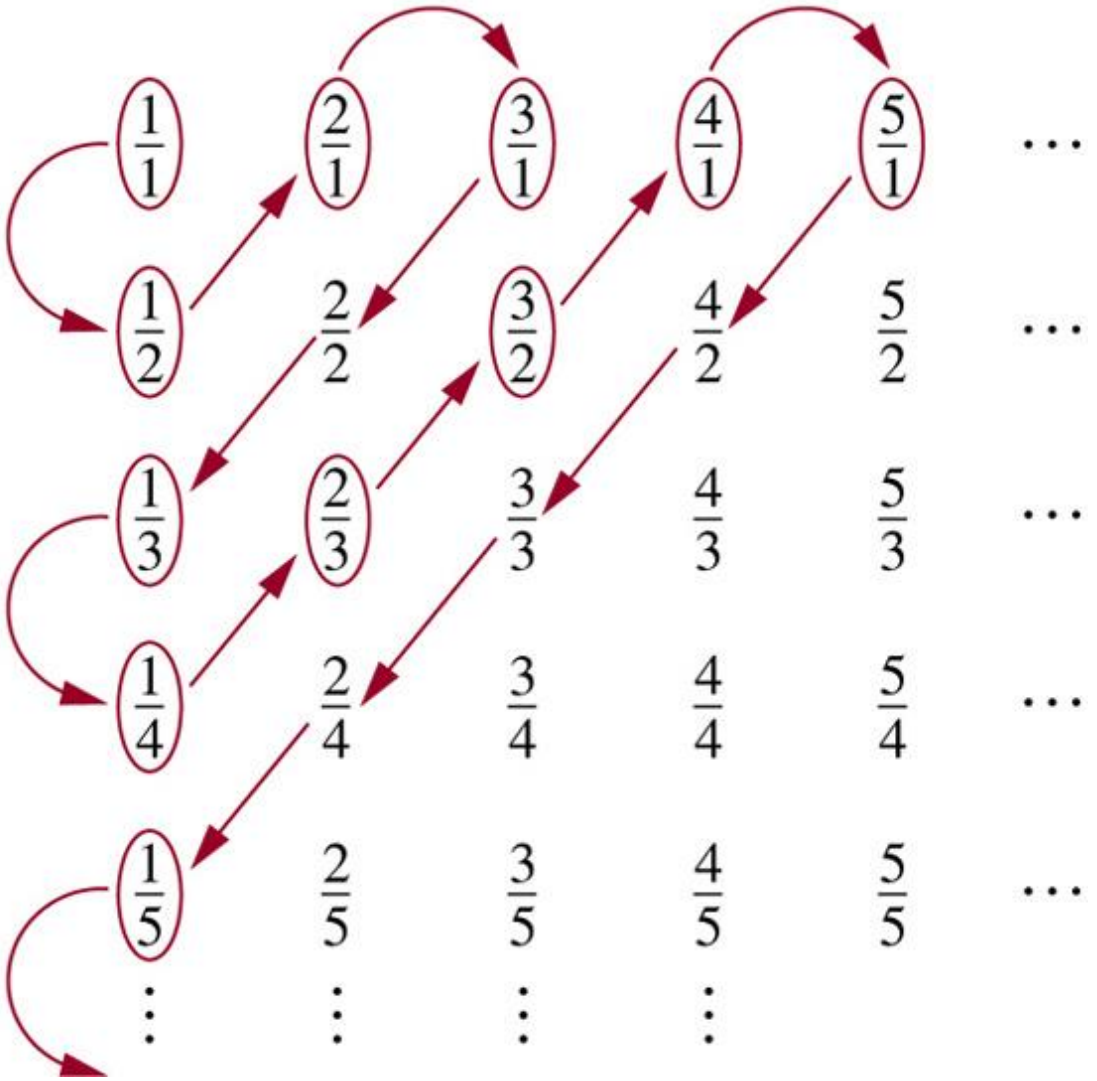
Cardinality of Infinite Sets

- The set of positive rational numbers is countable.
- To prove this we set up the postive rational numbers in a 2-D matrix in which the numerators increase as you move to the right in a row, and the denominators increase as you go down a column.
- You count them by moving along the diagonals of the matrix, skipping any rational numbers that we have already counted previously.

Cardinality of Infinite Sets

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Terms not circled are not listed because they repeat previously listed terms



Cardinality of Infinite Sets

• Can we establish a one-to-one correspondence between Z^+ and R, the set of real numbers?

• No. Goerg Cantor showed that we can't, using the Cantor diagonalization proof.

• Basically, this proof assumes that we can, and shows that this implies that we can put all the real numbers into a sequence, in which we can specify for any real number *x* what the "next" real number *y* is.

• However, for any two real numbers, no matter how close they are, we can always find another real number between them. Thus, we have a contradiction.

Cardinality of Infinite Sets

• We denote the cardinality of any infinite countable set by the symbol $\aleph_0$, pronounced "aleph null".

• We denote the cardinality of any uncountable set by the symbol $\aleph_1$, pronounced "aleph one".

Cardinality of Infinite Sets

- Is the set of all possible computer programs countable?
- Any given computer program can be represented in binary form, as a finite sequence of 0's and 1's.
- Any finite sequence of 0's and 1's can be interpreted as an integer.
- The set of integers is countable.
- Therefore, the set of all possible computer programs is countable.

Cardinality of Infinite Sets

- Is the set of all functions countable?
- Let us assume that each different function returns a different subset of R, the set of real numbers.
- There are $2^{|R|}$ different subsets of R. Since R is uncountable, $2^{|R|}$ is certainly also uncountable.
- So the set of all functions is uncountable.
- But the set of computer programs is countable.
- So there are fewer computer programs than there are functions that we might want to compute.

Conclusion

- In this chapter we have covered:
  - Introduction to sets
  - Set operations
  - Functions
  - Sequences and summations
  - Cardinality of infinite sets

Conclusion

- In this chapter we have covered:
  - Introduction to sets
  - Set operations
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